

5 (a) $T = 4 \text{ (ms) or } 4 \times 10^{-3} \text{ (s)}$ C1

$$f = 1/T = 1/0.004$$

$$= 250 \text{ Hz}$$
 A1 [2]

(b) intensity $\propto$ (amplitude)² and amplitude = 2.8 (2.83) (cm) B1

curve with same period and with amplitude 2.8 cm B1

curve shifted 1.0 ms to left or to right of wave X B1 [3]

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- (c) (i) gradient = $(4.5 - 2.4) \times 10^{-3} / (3.25 - 1.75) [= 1.4 \times 10^{-3}]$ B1
- wavelength = $0.45 \times 10^{-3} \times 1.4 \times 10^{-3}$ C1
- = $6.30 \times 10^{-7} \text{ (m)}$ C1
- = 630 nm A1 [4]
- (ii) (gradient is equal to λ/a therefore) gradient of line is reduced B1
- value of x will be reduced for all values of D
 or new line is completely below old line
 or intercept is less B1 [2]